PF-Lab 2 Assignment

Name:-Zain Mehdi Date:-9/10/2026

RollNo:13 BSCS Section A

LAB Task-1

A:- 9nput cpp

```
# include <iostream>
using namespace std;
int main(){
char letter;
letter = 'A';
cout <<letter<<'\n';
letter ='B';
cout letter '\n';
cin >> letter;
return 0;
}
```

Output

A
B

B:- 9nput cpp

```
#include <iostream>
using name space std;
int main (){
char letter;
letter = 65;
cout << letter<<'\n';
letter = 66;
cout << letter<<'\n';
cin >>letter;
return 0;
}
```

Output

A
B

CamScanner

Task:2

Source Code (cpp)

```
#include <iostream>
using namespace std;
int main(){
int a=30;
int b=10;
cout<<a+b<<'\n';
cout<<a-b<<'\n';
cout<<a*b<<'\n';
cout<<a/b<<'\n';
}
```

Output

```
40
20
300
3
```

```
return 0;
}
```

Task 3:-

Source Code (cpp)

```
#include <iostream>
using namespace std;
int main() {
int val1;
int val2;
int val3;
cout <<"Please enter your 3numbers";
cin >>val1 >> val2 >> val3;
cout<<val1 <<"\n" << val2 <<"\n" <<val3 <<"\n";
cout<<val3 <<"\n" <<val2 <<"\n" <<val1 <<"\n";
 return 0;
}
```

Output

```
Please Enter your 3numbers:
4 5 6
4
5
6
6
5
4
```

CamScanner

Task:4

```
#include <iostream>
using namespace std;
int main ( ){
int x1 = 5;
int x2 = 10;
int y1 = 15;
int y2 = 20;
cout << "Calculat the formula";
d= (x2-x1) *(x2-x1) + (y2-y1) *(y2-y1);
cout << d;
return 0;
}
```

Output

Calculate the formula 50

CamScanner

Task 5:-

1) The Area of Circle

```
#include <iostream>
using namespace std;
int main() {
int r = 5;
double area;
area = 3.14159 * r * r;
cout << "Area of Circle" << area;
}
```

Output

Area of Circle 78.5397

2) Area of Rectangle

```
#include <iostream>
using namespace std;
int main() {
int length = 7;
int width = 12;
double area;
area = length * width;
cout << "Area of Rectangle" <<
<< area;
}
```

Output

Area of Rectangle 84

3) Area of Triangle

```
#include <iostream>
using namespace std;
int main() {
int base = 7;
double height = 12.15152;
double area;
area = base * height * 0.5;
cout << "Area of Triangle"
<< area;
}
```

Output

Area of Triangle 42.5302

Task-6:

Source Code (cpp)

```
#include <iostream>
using namespace std;
int main( ) {
double p1=12.95, p2=24.95, p3=6.95, p4=14.95, p5=3.95;
double subtotal = p1+p2+p3+p4+p5;
double tax = subtotal x0.06;
double total= subtotal + tax;
cout << "price of item 1 = $" <<p1 <<endl;
cout << "price of item 2=$" <<p2<< endl;
cout <<"price of item 3=$" << p3<<endl;
cout <<"price of item 4=$" << p4 <<endl;
cout <<"price of item 5=$" << p5 <<endl;
cout<<"\n subtotal=$" << subtotal<< endl;
cout <<"sales tax (6%)=$" << tax <<endl;
cout <<"Total=$" << total << endl;
return 0;
}
```

Output

```
Price of item 1 = $12.95
Price of item 2 = $24.95
Price of item 3 = $ 6.95
Price of item 4 = $ 14.95
Price of item 5 = $ 3.95
   Subtotal = $63.75
Sales tax(6%) = $3.825
Total = $67.575
```

CamScanner

Bonus Task

Source Code (cpp)

```
#include <iostream>
using namespace std;
int main ( ) {
int a=10;
 int b=20;
 int c=30;
 int d=40;
 int e=50;
cout<<a<<"\t" << b<<"\t" << c<<"\t"
<<d<<"\t" <<e<<"\t";
}
```

Output

10 20 30 40 50

CamScanner

Home Tasks:-

Task:1

```
#include <iostream>
using namespace std;
int main(){
int var1,var2,temp;
cout<<"Enter First Number:";
cin>>var1;
cout<<"Enter Second Number:";
cin>>var2;
temp=var1;
var1=var2;
var2=temp;
cout<<"After Swapping:"<<endl;
cout<<"var1="<<var1<<endl;
cout<<"var2="<<var2<<endl;
return 0;
}
```

Output:-

```
Enter first Number:5
Enter Second Number:10
After swapping:
var1=10
var2=5
```

Task:2

Source Code (cpp)

```
#include <iostream>
using namespace std;
int main ( ) {
char ch;
cout<<"Enter a small alphabet:";
cin>> ch;
ch=ch-32;
cout <<"Uppercase:" <<ch;
return 0;
}
```

Output

```
Enter a small alphabet:z
Uppercase:Z
```

CamScanner

C:\Users\PERFECT PC\Desktop\Books\Task 1 Again.cpp - Dev-C++ 5.11

File Edit Search View Project Execute Tools AStyle Window Help

TDM-GCC 4.9.2 64-bit Release

(globals)

Task 1 Again.cpp

```cpp
#include <iostream>
using namespace std;
int main(){
    char letter;
    letter = 'A';
    cout << letter << '\n';
    letter = 'B';
    cout << letter << '\n';
    cin >> letter;
    return 0;
}
```

Compiler Resources Compile Log Debug Find Results Close

Abort Compilation

Shorten compiler paths

```
Compilation results...
--------
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\PERFECT PC\Desktop\Books\Task 1 Again.exe
- Output Size: 1.83190059661865 MiB
- Compilation Time: 1.27s
```

Line: 11 Col: 2 Sel: 0 Lines: 11 Length: 209 Insert Done parsing in 0.687 seconds

Type here to search

Hot...

5:40 AM
10/9/2026

CamScanner



CamScanner

C:\Users\PERFECT PC\Desktop\Books\Task 1 Again.cpp - [Executing] - Dev-C++ 5.11

File Edit Search View Project Execute Tools AStyle Window Help

TDM-GCC 4.9.2 64-bit Release

(globals)

[*] Task 1 Again.cpp

```
1   #include <iostream>
2   using namespace std;
3   int main(){
4       char letter;
5       letter = 65;
6       cout << letter << '\n';
7       letter = 66;
8       cout << letter << '\n';
9       cin >> letter;
10      return 0;
11  }
```

Compiler Resources Compile Log Debug Find Results Close

Abort Compilation

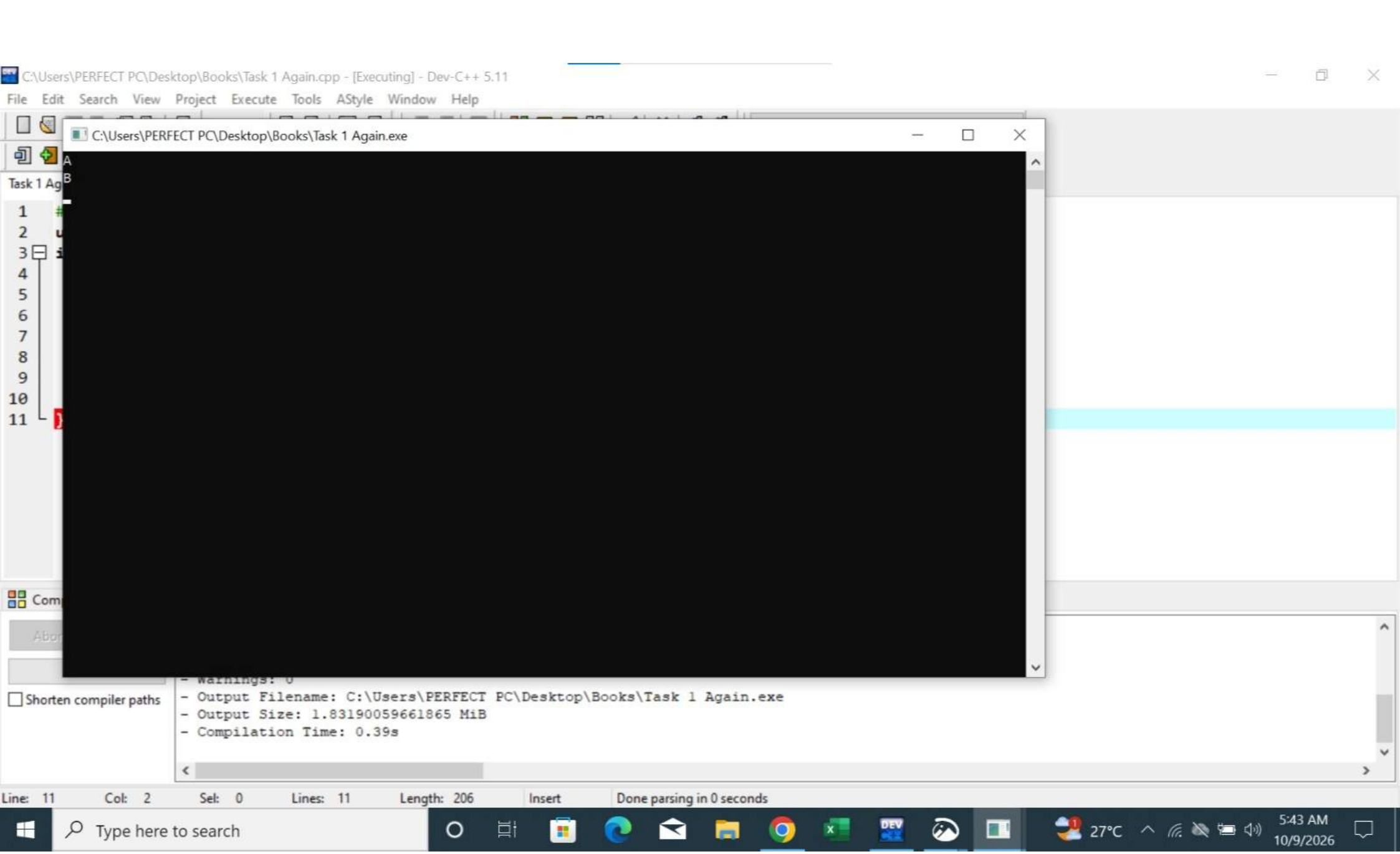
Shorten compiler paths

```
Compilation results...
--------
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\PERFECT PC\Desktop\Books\Task 1 Again.exe
- Output Size: 1.83190059661865 MiB
- Compilation Time: 0.39s
```

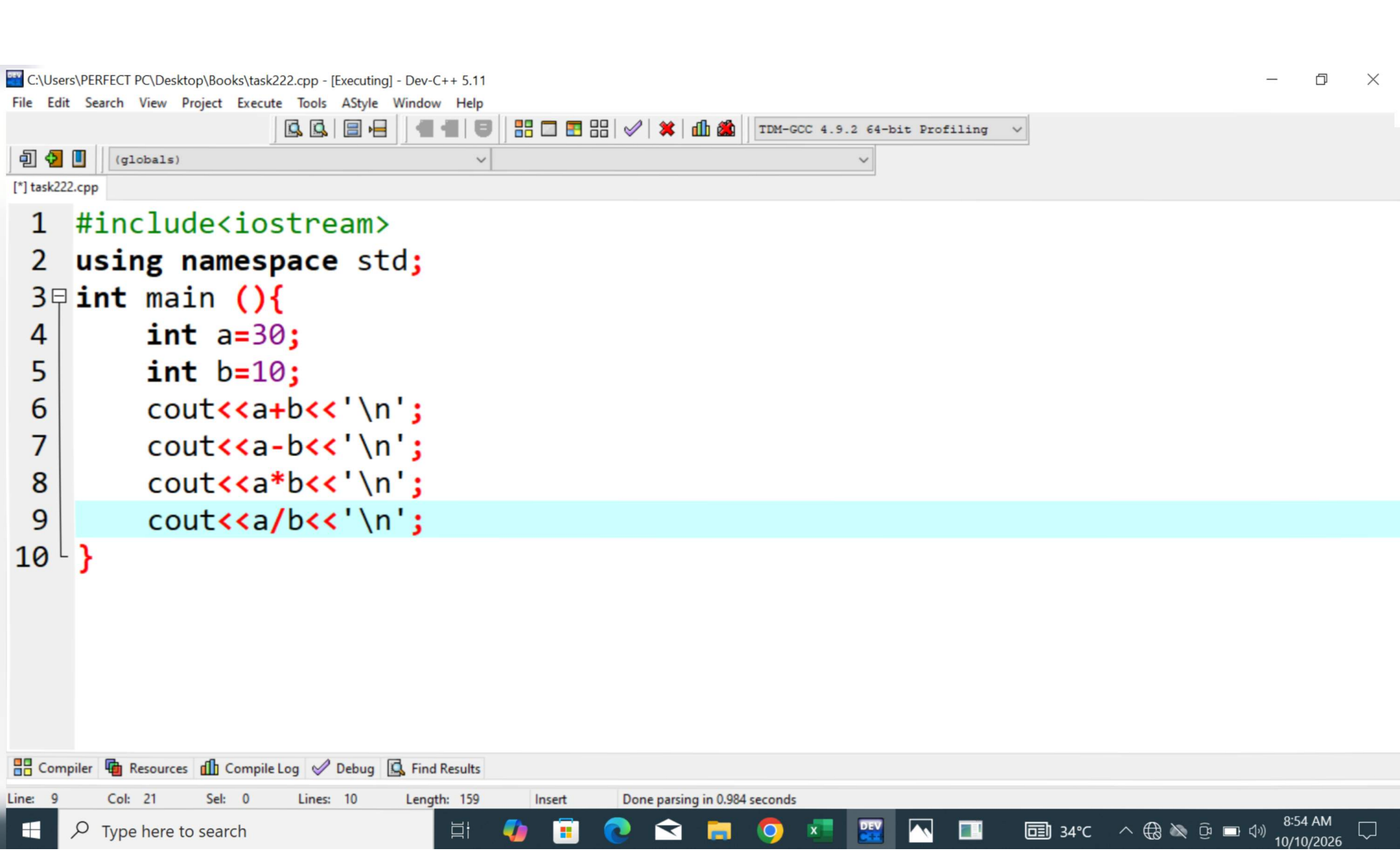
Line: 11 Col: 2 Sel: 0 Lines: 11 Length: 206 Insert Done parsing in 0.687 seconds

Type here to search 27°C 5:42 AM 10/9/2026

CamScanner


C:\Users\PERFECT PC\Desktop\Books\Task 1 Again.cpp - [Executing] - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
C:\Users\PERFECT PC\Desktop\Books\Task 1 Again.exe
A
B
- Output Filename: C:\Users\PERFECT PC\Desktop\Books\Task 1 Again.exe
- Output Size: 1.83190059661865 MiB
- Compilation Time: 0.39s
Shorten compiler paths
Line: 11 Col: 2 Sel: 0 Lines: 11 Length: 206 Insert Done parsing in 0 seconds
Type here to search
27°C
5:43 AM
10/9/2026

CamScanner


C:\Users\PERFECT PC\Desktop\Books\task222.cpp - [Executing] - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
TDM-GCC 4.9.2 64-bit Profiling
(globals)
[*] task222.cpp
#include<iostream>
using namespace std;
int main (){
int a=30;
int b=10;
cout<<a+b<<'\n';
cout<<a-b<<'\n';
cout<<a*b<<'\n';
cout<<a/b<<'\n';
}
Compiler Resources Compile Log Debug Find Results
Line: 9 Col: 21 Sel: 0 Lines: 10 Length: 159 Insert Done parsing in 0.984 seconds
Type here to search
34°C
8:54 AM
10/10/2026

CamScanner

Users\PERFECT PC\Desktop\Books\task222.cpp - [Executing] - Dev-C++ 5.11

C:\Users\PERFECT PC\Desktop\Books\task222.exe

```
40
20
300
3

--------------------------------
Process exited after 0.9815 seconds with return value 0
Press any key to continue . . .
```

CamScanner

C:\Users\PERFECT PC\Desktop\Books\task222.cpp - Dev-C++ 5.11

File Edit Search View Project Execute Tools AStyle Window Help

TDM-GCC 4.9.2 64-bit Profiling

(globals)

[*] task222.cpp

```cpp
#include<iostream>
using namespace std;
int main (){
    int val1;
    int val2;
    int val3;
    cout <<"Please enter you 3 numbers";
    cin>>val1>>val2>>val3;
    cout<<val1<<"\n"<<val2<<"\n"<<val3<<"\n";
    cout<<val3<<"\n"<<val2<<"\n"<<val1<<"\n";
    system("PAUSE");
    return 0;
}
```

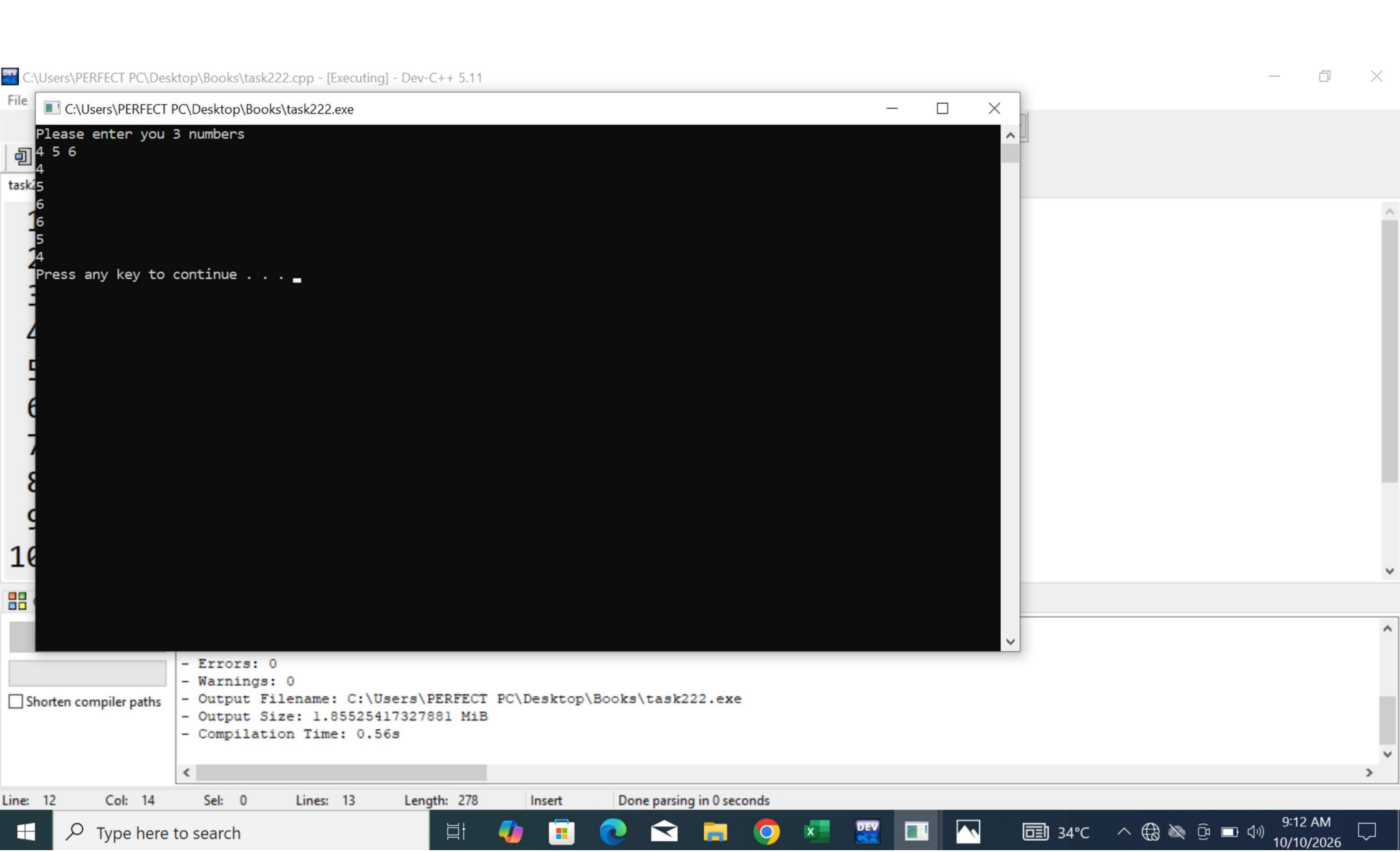
Compiler (2) Resources Compile Log Debug Find Results

Line: 12 Col: 14 Sel: 0 Lines: 13 Length: 278 Insert Done parsing in 0 seconds

Type here to search

9:11 AM
10/10/2026

CamScanner


C:\Users\PERFECT PC\Desktop\Books\task222.cpp - [Executing] - Dev-C++ 5.11
File
C:\Users\PERFECT PC\Desktop\Books\task222.exe
Please enter you 3 numbers
4 5 6
4
5
6
6
5
4
Press any key to continue . . .
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\PERFECT PC\Desktop\Books\task222.exe
- Output Size: 1.85525417327881 MiB
- Compilation Time: 0.56s
Shorten compiler paths
Line: 12 Col: 14 Sel: 0 Lines: 13 Length: 278 Insert Done parsing in 0 seconds
Type here to search
34°C
9:12 AM
10/10/2026

CamScanner

C:\Users\PERFECT PC\Desktop\Books\Untitled2.cpp - [Executing] - Dev-C++ 5.11

File Edit Search View Project Execute Tools AStyle Window Help

TDM-GCC 4.9.2 64-bit Profiling

(globals)

task222.cpp Untitled2.cpp

```
1  #include<iostream>
2  using namespace std;
3  int main (){
4      int x1=5;
5      int x2=10;
6      int y1=15;
7      int y2=20;
8      int d;
9      cout<<"Calculate the formula";
10     d=(x2-x1)*(x2-x1) + (y2-y1)*(y2-y1);
11     cout<<d;
12     return 0;
13 }
```

Compiler Resources Compile Log Debug Find Results

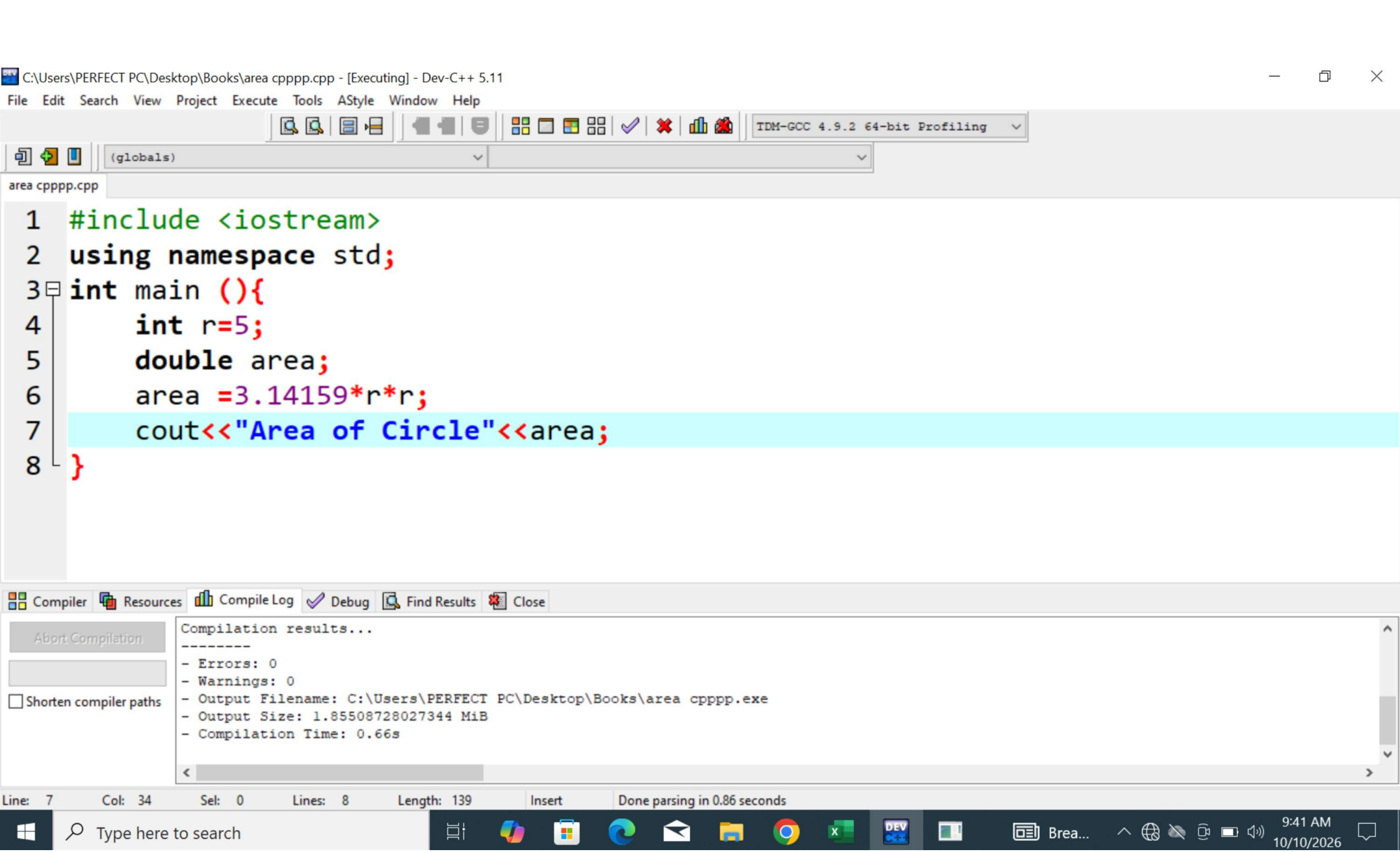
Line: 12 Col: 14 Sel: 0 Lines: 13 Length: 214 Insert Done parsing in 0.016 seconds

Type here to search 34°C 9:27 AM 10/10/2026

CamScanner

```
Calculate the formula50
--------------------------------
Process exited after 0.8342 seconds with return value 0
Press any key to continue . . .
```

CamScanner


C:\Users\PERFECT PC\Desktop\Books\area cpppp.cpp - [Executing] - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
TDM-GCC 4.9.2 64-bit Profiling
(globals)
area cpppp.cpp
1 #include <iostream>
2 using namespace std;
3 int main (){
4 int r=5;
5 double area;
6 area =3.14159*r*r;
7 cout<<"Area of Circle"<<area;
8 }
Compiler Resources Compile Log Debug Find Results Close
Abort Compilation
Shorten compiler paths
Compilation results...

- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\PERFECT PC\Desktop\Books\area cpppp.exe
- Output Size: 1.85508728027344 MiB
- Compilation Time: 0.66s
Line: 7 Col: 34 Sel: 0 Lines: 8 Length: 139 Insert Done parsing in 0.86 seconds
Type here to search
9:41 AM
10/10/2026

CamScanner

```
rea of Circle78.5397
------------------------------
rocess exited after 0.9601 seconds with return value 0
ress any key to continue . . .
```

CamScanner

C:\Users\PERFECT PC\Desktop\Books\Area of Rectangle.cpp - Dev-C++ 5.11

File Edit Search View Project Execute Tools AStyle Window Help

TDM-GCC 4.9.2 64-bit Profiling

(globals)

area cpppp.cpp | [*] Area of Rectangle.cpp

```cpp
#include <iostream>
using namespace std;
int main (){
    int length=7;
    int width=12;
    double area;
    area =length*width;
    cout<<"Area of Rectangle"<<area;
}
```

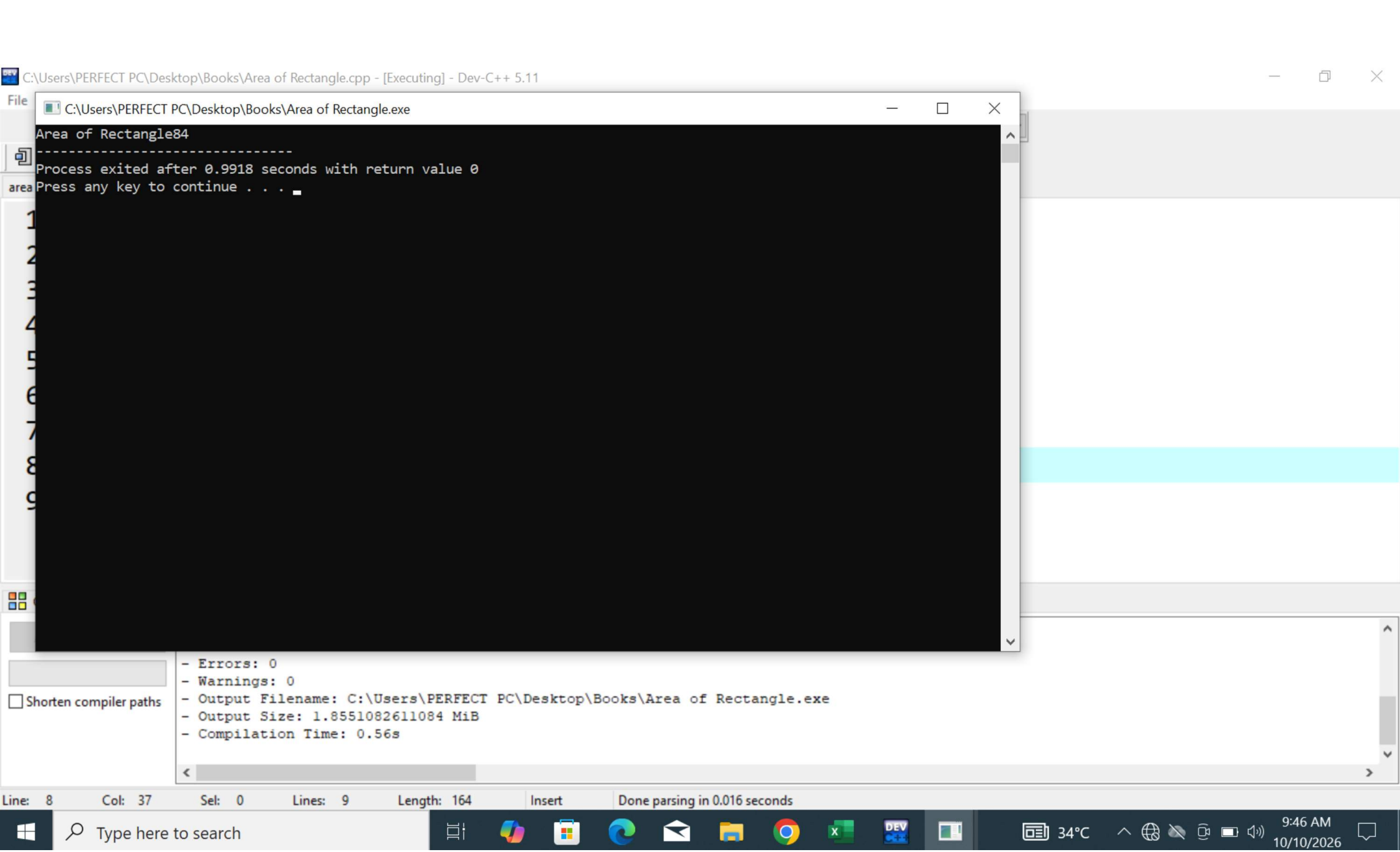
Compiler (3) | Resources | Compile Log | Debug | Find Results | Close

Line	Col	File	Message
1	2	C:\Users\PERFECT PC\Desktop\Books\Area of Rectangle....	[Error] invalid preprocessing directive #inlude
		C:\Users\PERFECT PC\Desktop\Books\Area of Rectan...	**In function 'int main()':**
8	2	C:\Users\PERFECT PC\Desktop\Books\Area of Rectangle....	[Error] 'cout' was not declared in this scope

Line: 8 Col: 37 Sel: 0 Lines: 9 Length: 164 Insert Done parsing in 0.031 seconds

Type here to search 34°C 9:46 AM 10/10/2026

CamScanner


C:\Users\PERFECT PC\Desktop\Books\Area of Rectangle.cpp - [Executing] - Dev-C++ 5.11
File
C:\Users\PERFECT PC\Desktop\Books\Area of Rectangle.exe
Area of Rectangle84
Process exited after 0.9918 seconds with return value 0
Press any key to continue . . .
area
Shorten compiler paths
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\PERFECT PC\Desktop\Books\Area of Rectangle.exe
- Output Size: 1.8551082611084 MiB
- Compilation Time: 0.56s
Line: 8 Col: 37 Sel: 0 Lines: 9 Length: 164 Insert Done parsing in 0.016 seconds
Type here to search
34°C
9:46 AM
10/10/2026

CamScanner

C:\Users\PERFECT PC\Desktop\Books\Area of triangle.cpp - [Executing] - Dev-C++ 5.11

File Edit Search View Project Execute Tools AStyle Window Help

TDM-GCC 4.9.2 64-bit Profiling

(globals)

area cpppp.cpp | Area of Rectangle.cpp | Area of triangle.cpp

```cpp
#include <iostream>
using namespace std;
int main (){
    int base=7;
    double height=12.15152;
    double area;
    area=base*height*0.5;
    cout<<"Area of Triangle"<<area;
}
```

Compiler | Resources | Compile Log | Debug | Find Results | Close

Abort Compilation

Shorten compiler paths

```
Compilation results...
--------
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\PERFECT PC\Desktop\Books\Area of triangle.exe
- Output Size: 1.85510730743408 MiB
- Compilation Time: 0.63s
```

Line: 8 Col: 36 Sel: 0 Lines: 9 Length: 173 Insert Done parsing in 0.032 seconds

Type here to search

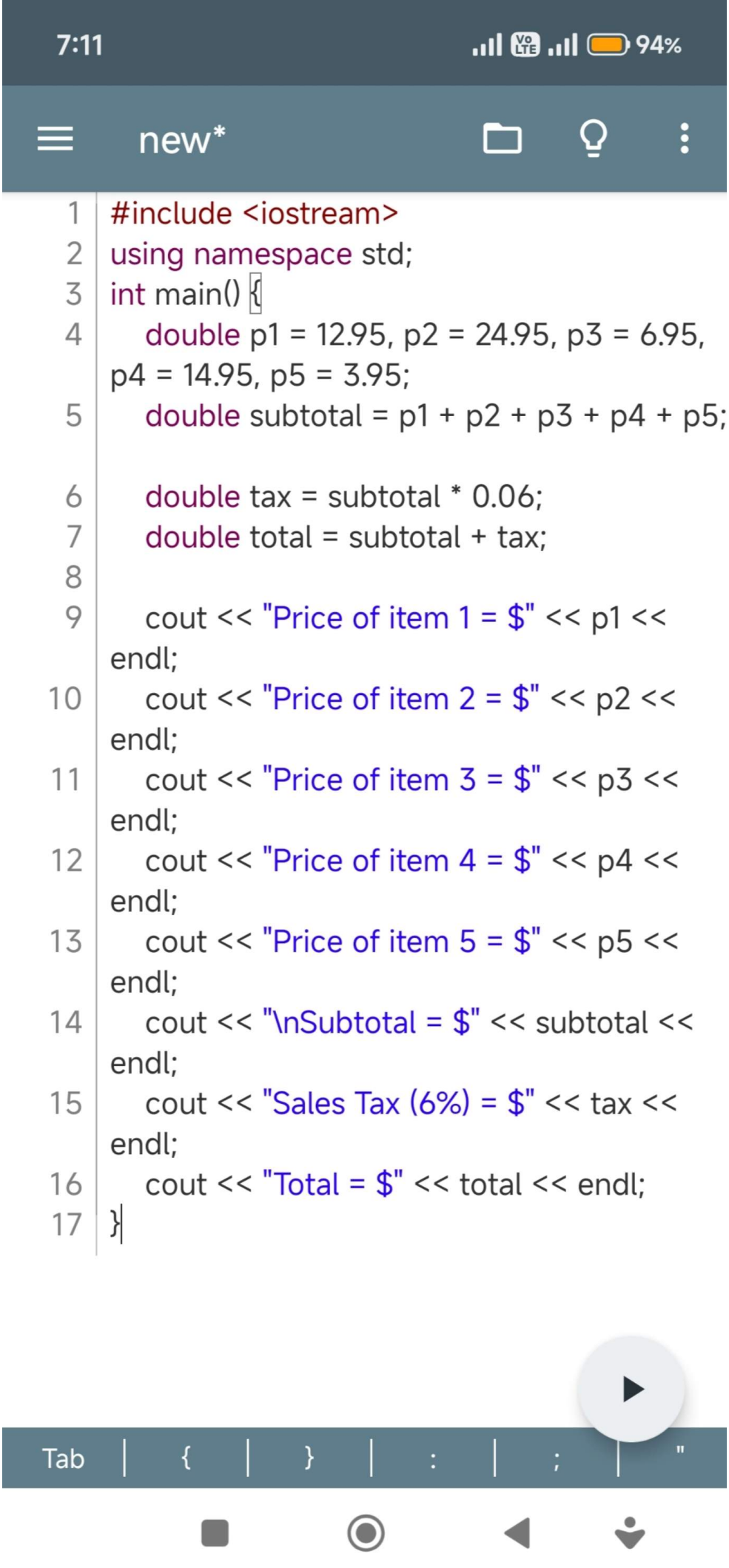
34°C

9:52 AM
10/10/2026

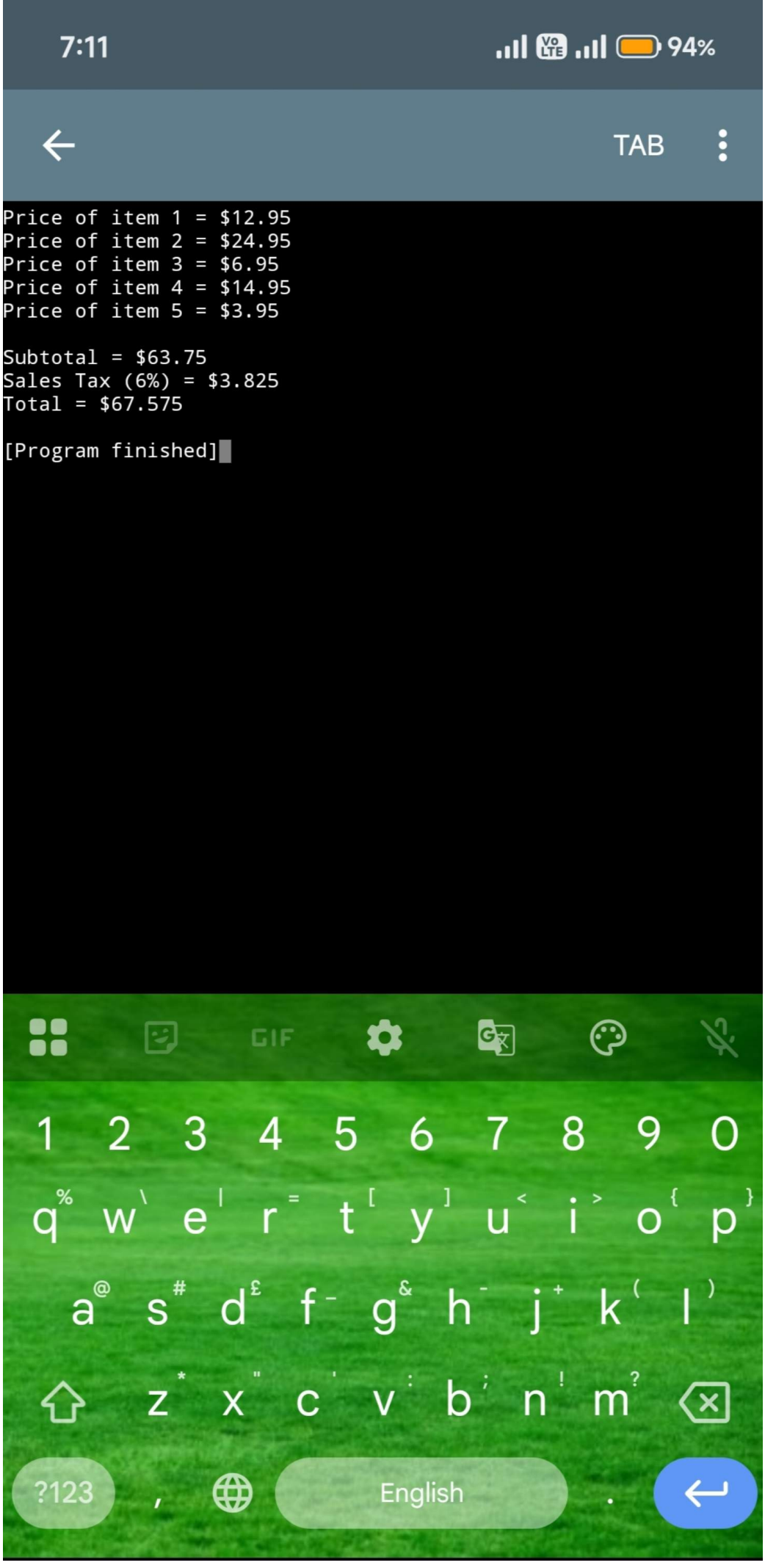
CamScanner

```
Area of Triangle42.5303
--------------------------------
Process exited after 1.002 seconds with return value 0
Press any key to continue . . .
```

CamScanner


7:11
94%
new*
1 #include <iostream>
2 using namespace std;
3 int main() {
4 double p1 = 12.95, p2 = 24.95, p3 = 6.95, p4 = 14.95, p5 = 3.95;
5 double subtotal = p1 + p2 + p3 + p4 + p5;
6 double tax = subtotal * 0.06;
7 double total = subtotal + tax;
8
9 cout << "Price of item 1 = $" << p1 << endl;
10 cout << "Price of item 2 = $" << p2 << endl;
11 cout << "Price of item 3 = $" << p3 << endl;
12 cout << "Price of item 4 = $" << p4 << endl;
13 cout << "Price of item 5 = $" << p5 << endl;
14 cout << "\nSubtotal = $" << subtotal << endl;
15 cout << "Sales Tax (6%) = $" << tax << endl;
16 cout << "Total = $" << total << endl;
17 }
Tab
{
}
:
;
"

CamScanner


7:11
94%
TAB
Price of item 1 = $12.95
Price of item 2 = $24.95
Price of item 3 = $6.95
Price of item 4 = $14.95
Price of item 5 = $3.95
Subtotal = $63.75
Sales Tax (6%) = $3.825
Total = $67.575
[Program finished]
GIF
1 2 3 4 5 6 7 8 9 0
q w e r t y u i o p
a s d f g h j k l
z x c v b n m
?123
English

CamScanner

C:\Users\PERFECT PC\Desktop\Books\Untitled4.cpp - [Executing] - Dev-C++ 5.11

File Edit Search View Project Execute Tools AStyle Window Help

TDM-GCC 4.9.2 64-bit Profiling

(globals)

area cpppp.cpp | Area of Rectangle.cpp | Area of triangle.cpp | Untitled4.cpp

```cpp
#include <iostream>
using namespace std;
int main (){
    int a=10;
    int b=20;
    int c=30;
    int d=40;
    int e=50;
    cout<<a<<"\t"<<b<<"\t"<<c<<"\t"<<d<<"\t"<<e<<"\t";
}
```

Compiler | Resources | Compile Log | Debug | Find Results | Close

Abort Compilation

Shorten compiler paths

```
Compilation results...
--------
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\PERFECT PC\Desktop\Books\Untitled4.exe
- Output Size: 1.85508728027344 MiB
- Compilation Time: 0.55s
```

Line: 9 Col: 50 Sel: 0 Lines: 10 Length: 173 Insert Done parsing in 0.016 seconds

Type here to search

Hot...

10:07 AM
10/10/2026

CamScanner

```
10      20      30      40      50
--------------------------------
Process exited after 0.9754 seconds with return value 0
Press any key to continue . . .
```

CamScanner

C:\Users\PERFECT PC\Desktop\Books\Untitled5.cpp - [Executing] - Dev-C++ 5.11

File Edit Search View Project Execute Tools AStyle Window Help

TDM-GCC 4.9.2 64-bit Profiling

(globals)

area cpppp.cpp | Area of Rectangle.cpp | Area of triangle.cpp | [*] Untitled4.cpp | Untitled5.cpp

```cpp
#include <iostream>
using namespace std;
int main (){
    int var1,var2,temp;
    cout<<"Enter First Number:";
    cin>>var1;
    cout<<"Enter Second Number:";
    cin>>var2;
    temp=var1;
    var1=var2;
    var2=temp;
    cout<<"After swapping:"<<endl;
    cout<<"var1="<<var1<<endl;
    cout<<"var2="<<var2<<endl;
    return 0;
}
```

Compiler | Resources | Compile Log | Debug | Find Results

Line: 15 Col: 14 Sel: 0 Lines: 16 Length: 313 Insert Done parsing in 0.016 seconds

Type here to search 34°C 10:19 AM 10/10/2026

CamScanner

C:\Users\PERFECT PC\Desktop\Books\Untitled5.exe

```
Enter First Number:5
Enter Second Number:10
After swapping:
var1=10
var2=5

--------------------------------
Process exited after 14.23 seconds with return value 0
Press any key to continue . . .
```

CamScanner

C:\Users\PERFECT PC\Desktop\Books\Untitled6.cpp - [Executing] - Dev-C++ 5.11

File Edit Search View Project Execute Tools AStyle Window Help

TDM-GCC 4.9.2 64-bit Profiling

(globals)

area cpppp.cpp | Area of Rectangle.cpp | Area of triangle.cpp | [*] Untitled4.cpp | Untitled5.cpp | Untitled6.cpp

```cpp
#include<iostream>
using namespace std;
int main (){
    char ch;
    cout<<"Enter a small alphabet:";
    cin>>ch;
    ch=ch-32;
    cout<<"Uppercase:"<<ch;
    return 0;
}
```

Compiler | Resources | Compile Log | Debug | Find Results | Close

Abort Compilation

Shorten compiler paths

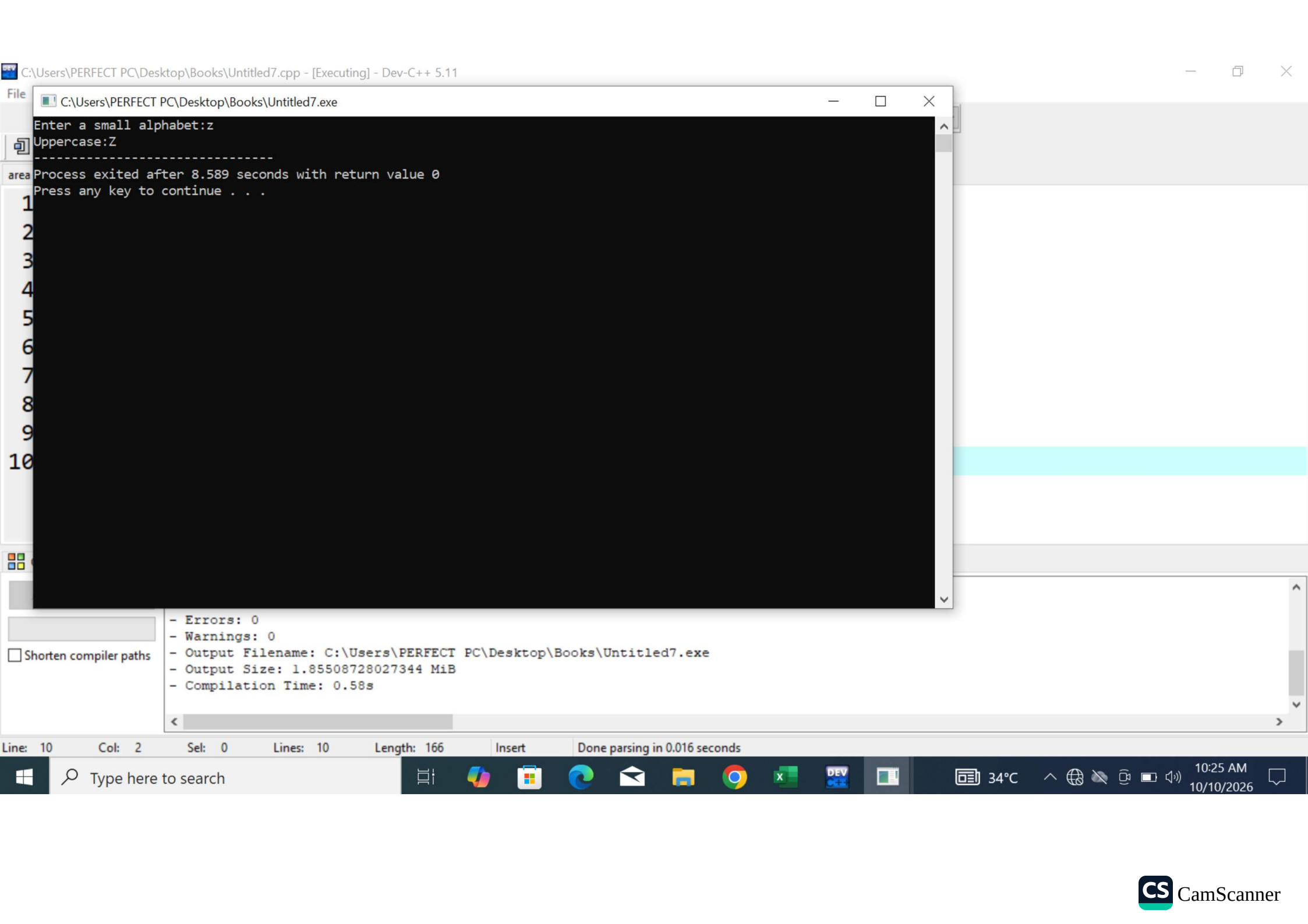
```
Compilation results...
--------
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\PERFECT PC\Desktop\Books\Untitled6.exe
- Output Size: 1.85508728027344 MiB
- Compilation Time: 0.66s
```

Line: 9 Col: 14 Sel: 0 Lines: 10 Length: 166 Insert Done parsing in 0.016 seconds

Type here to search

34°C 10:24 AM 10/10/2026

CamScanner


C:\Users\PERFECT PC\Desktop\Books\Untitled7.cpp - [Executing] - Dev-C++ 5.11
File
C:\Users\PERFECT PC\Desktop\Books\Untitled7.exe
Enter a small alphabet:z
Uppercase:Z
Process exited after 8.589 seconds with return value 0
Press any key to continue . . .
area
Shorten compiler paths
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\PERFECT PC\Desktop\Books\Untitled7.exe
- Output Size: 1.85508728027344 MiB
- Compilation Time: 0.58s
Line: 10 Col: 2 Sel: 0 Lines: 10 Length: 166 Insert Done parsing in 0.016 seconds
Type here to search
34°C
10:25 AM
10/10/2026

CamScanner